the mean free time between collisions. This may be shown as follows: Let $P(t)$ be the probability that t seconds after a certain collision has occurred, an electron has not yet collided again; $P(t + dt)$ represents the same quantity after $(t + dt)$ seconds. Then

$$P(t + dt) = P(t) + (dP/dt)\, dt$$

On the other hand, we may also write

$$P(t + dt) = P(t)P\,(dt) = P(t)(1 - dt/\tau)$$

where $(1 - dt/\tau)$ represents the probability for an electron not to collide during the interval dt. From the last two equations one finds

$$P(t) = e^{-t/\tau}$$

since $P = 1$ for $t = 0$. Hence the mean free time between collisions is

$$\langle t \rangle = \int_0^\infty t\,(dP/dt)\, dt = \tau \tag{11-10}$$

It must be emphasized, however, that the relaxation time and the mean free time between collisions are identical only if the velocity after collision is random. For example, if the scattering is not isotropic and τ_c is the mean free time between collisions, the relaxation time can readily be shown to be

$$\tau = \tau_c/(1 - \langle \cos \beta \rangle) \tag{11-11}$$

where $\langle \cos \beta \rangle$ is the average of the cosine of the scattering angle.[2] Thus when nearly all collisions involve small angles, the electron has a rather strong "memory" and it takes a relatively large number of collisions to erase this memory, i.e., $\tau \gg \tau_c$ in that case.

11-3. The Boltzmann transport equation

It will be evident that in a state of steady flow of heat or electricity, the distribution function for the velocity components and spatial coordinates of the electrons will be different from that in thermal equilibrium in the absence of flow. Thus the theory of transport phenomena is concerned with determining this distribution function for given external fields. We shall see in this section that the determination of the distribution function requires solving an integrodifferential equation, viz., the Boltzmann transport equation.[3]

[2] See, for example, W. Shockley, *Electrons and Holes in Semiconductors*, Van Nostrand, New York, 1950, p. 255.

[3] L. Boltzmann, *Vorlesungen über Gastheorie*, Barth, Leipzig, 1923.

Let p_x, p_y, p_z represent the components of the momentum of an electron and let

$$f(p_x p_y p_z ; xyz ; t)\, dp_x\, dp_y\, dp_z\, dx\, dy\, dz \tag{11-12}$$

represent the number of electrons in the volume element $dx\, dy\, dz$ which at the instant t have momenta in the range $dp_x\, dp_y\, dp_z$. The steady state is then defined by

$$\delta f / \delta t = 0 \tag{11-13}$$

In order to obtain information about the function f, it is necessary to consider the causes which, when operative by themselves, would tend to produce a change of f with time. First of all, we must consider the rate of change of f resulting from the velocities of the electrons and from the components X, Y, Z of the external forces which are assumed to act on the electrons. Consider the group of particles defined by (11-12) at an instant $t + \delta t$, where δt is a very small time interval. The momenta and spatial coordinates of this group of electrons at $t + \delta t$ are then to be found about the point

$$\begin{aligned} &p_x + X\,\delta t; \quad p_y + Y\,\delta t; \quad p_z + Z\,\delta t \\ &x + p_x\,\delta t/m; \quad y + p_y\,\delta t/m; \quad z + p_z\,\delta t/m \end{aligned} \tag{11-14}$$

However, according to the definition of the distribution function (11-12) the number of electrons which at the instant $t + \delta t$ have their representative points in an element $dp_x\, dp_y\, dp_z\, dx\, dy\, dz$ around the point defined by (11-14) must be equal to

$$f(p_x + X\,dt, \ldots; x + p_x\,\delta t/m \ldots; t + \delta t)\, dp_x\, dp_y\, dp_z\, dx\, dy\, dz \tag{11-15}$$

Since (11-12) and (11-15) must be equal, one is led to the following result, obtained by expanding (11-15),

$$(\partial f/\partial t)_{\text{fields}} = -\frac{\partial f}{\partial p_x} X - \frac{\partial f}{\partial p_y} Y - \frac{\partial f}{\partial p_z} Z - \frac{\partial f}{\partial x} v_x - \frac{\partial f}{\partial y} v_y - \frac{\partial f}{\partial z} v_z \tag{11-16}$$

where v_x, v_y, v_z represent the velocity components. In the steady state there must be other processes which just balance the rate of change (11-16) produced by fields and gradients. As we noted already in the preceding section, such processes are provided by electron-lattice interactions. Thus condition (11-13) may be written in the form

$$(\partial f/\partial t)_{\text{fields}} + (\partial f/\partial t)_{\text{coll}} = 0 \tag{11-17}$$

where the first term is given by (11-16) and where the second term refers to electron-lattice scattering (compare 11-6). Since the force exerted on

an electron by a combined electric and magnetic field is given by the Lorentz expression

$$\boldsymbol{F} = -e\left(\boldsymbol{E} + \frac{1}{c}\,\boldsymbol{v} \times \boldsymbol{H}\right)$$

where $\boldsymbol{v}$ is the velocity vector, we may write (11-17) combined with (11-16) in the general form

$$(\partial f/\partial t)_{\text{coll}} = -e\left(\boldsymbol{E} + \frac{1}{c}\,\boldsymbol{v} \times \boldsymbol{H}\right) \cdot \text{grad}_{\boldsymbol{p}} f + \boldsymbol{v} \cdot \text{grad}_{\boldsymbol{r}} f \qquad (11\text{-}18)$$

which is the Boltzmann transport equation for electrons.

The left hand side of this equation involves an integral operator, making the equation an integrodifferential equation; this may be seen as follows: The number of electrons per unit volume which, due to collisions with the lattice, change their momenta per unit time from the range $dp_x\,dp_y\,dp_z$ to another range $dp'_x\,dp'_y\,dp'_z$ can be represented by

$$f(\boldsymbol{p},\boldsymbol{r},t)\,dp_x\,dp_y\,dp_z\,P(\boldsymbol{p},\boldsymbol{p}',\boldsymbol{r})\,dp'_x\,dp'_y\,dp'_z$$

where the transition probability $P(\boldsymbol{p},\boldsymbol{p}',\boldsymbol{r})$ is determined by the type of electron-lattice interaction. Similarly, the corresponding number of electrons thrown from the range $dp'_x\,dp'_y\,dp'_z$ into $dp_x\,dp_y\,dp_z$ per unit time is

$$f(\boldsymbol{p}',\boldsymbol{r},t)\,dp'_x\,dp'_y\,dp'_z\,P(\boldsymbol{p}',\boldsymbol{p},\boldsymbol{r})\,dp_x\,dp_y\,dp_z$$

The net difference between the above quantities integrated over $dp'_x\,dp'_y\,dp'_z$ determines $(\partial f/\partial t)_{\text{coll}}$, i.e.,

$$(\partial f/\partial t)_{\text{coll}} = \iiint [f(\boldsymbol{p}',\boldsymbol{r},t)P(\boldsymbol{p}',\boldsymbol{p},\boldsymbol{r}) - f(\boldsymbol{p},\boldsymbol{r},t)P(\boldsymbol{p},\boldsymbol{p}',\boldsymbol{r})]\,dp'_x\,dp'_y\,dp'_z \qquad (11\text{-}19)$$

It is evident that since the left-hand side of (11-18) is given by expression (11-19), containing the transition probabilities P, the distribution function in the state of steady flow depends explicitly on the mechanism of interaction between the electrons and the lattice. From the atomic theory of electron scattering it can be shown that under certain circumstances it is possible to define a relaxation time such that $(\partial f/\partial t)_{\text{coll}}$ takes the form

$$(\partial f/\partial t)_{\text{coll}} = -\frac{f(\boldsymbol{p},\boldsymbol{r}) - f_0}{\tau(\boldsymbol{p},\boldsymbol{r})} \qquad (11\text{-}20)$$

(compare 11-5). Here f_0 represents the distribution function in thermal equilibrium in the absence of external fields. The physical meaning of the relaxation time $\tau(\boldsymbol{p},\boldsymbol{r})$ is analogous to that of τ introduced in the preceding section: when the external fields are suddenly removed, $(f - f_0)$ decays to zero in the fashion

$$(f - f_0)_t = (f - f_0)_{t=0} e^{-t/\tau} \qquad (11\text{-}21)$$

When for a certain problem a relaxation time exists, the treatment is strongly simplified, since the integrodifferential equation then becomes an ordinary equation. An example of this type will be discussed in the next section.

Special cases for which a relaxation time can be defined consistently may be mentioned here.[4]

(i) In processes whereby the electrons may be considered to be scattered by elastic spheres; this is of importance for that part of the resistivity which is due to impurity scattering.

(ii) From the (approximate) theory of the interaction between electrons and lattice vibrations it follows that a relaxation time can be defined when $(\theta/T)^2 \ll 1$, where θ is the Debye temperature. This simplifies the theory of electrical and thermal conductivities at high temperatures.

11-4. The Sommerfeld theory of electrical conductivity

A theory of metallic conductivity based on average velocities, as employed in Sec. 11-2, was developed by Drude in 1900. Lorentz in 1905 reinvestigated the problem, using the Boltzmann transport equation and a simplified model for the collisions between the electrons and atoms in the lattice. However, the use of classical statistics led to serious difficulties; for a review of these theories we refer to the literature.[4] In 1928 Sommerfeld recalculated the conductivities along the lines of Lorentz' theory, but replacing classical statistics by Fermi-Dirac statistics.[5] Sommerfeld did not investigate the actual mechanism of interaction between the electrons and the lattice any further, but assumed that a relaxation time can be defined which is a function of the energy of the electrons only. As an application to the Boltzmann transport equation we shall discuss below Sommerfeld's theory of the electrical conductivity based on the free electron approximation; the thermal conductivity will be discussed later.

The number of electronic states per unit volume associated with an element $dp_x\, dp_y\, dp_z$ in momentum space is (including the spin) $(2/h^3)\, dp_x\, dp_y\, dp_z$. In thermal equilibrium and in the absence of fields, let the average number of occupied states be

$$(2/h^3)F_0(p)\, dp_x\, dp_y\, dp_z$$

where F_0 is the Fermi distribution function in terms of the total momentum p. Suppose we apply an electric field E_x along the x-direction, other

[4] See, for example, A. H. Wilson, *The Theory of Metals*, 2d ed., Cambridge, London, 1953, pp. 8, 264.

[5] A. Sommerfeld, *Z. Physik*, **47**, 1 (1928); see also the article by A. Sommerfeld and H. Bethe, in *Handbuch der Physik*, Vol. 24/2, 1934, or A. Sommerfeld and N. H. Frank, *Revs. Mod. Phys.*, **3**, 1 (1931).

fields or gradients being absent. When in the state of steady current the average number of electrons per unit volume in the range $dp_x\,dp_y\,dp_z$ is represented by

$$(2/h^3)F(\boldsymbol{p})\,dp_x\,dp_y\,dp_z \tag{11-22}$$

we may write immediately for the current density,

$$I_x = -\frac{2e}{h^3}\iiint v_x(F - F_0)\,dp_x\,dp_y\,dp_z \tag{11-23}$$

which is a generalization of (11-3). The term F_0 does not contribute anything to the current, since it is spherically symmetric with respect to $\boldsymbol{p}$; it has been added, however, to emphasize the fact that the current is essentially determined by the deviation $(F - F_0)$ from the Fermi distribution. Thus, if one can calculate $(F - F_0)$, I_x may be obtained.

The Boltzmann transport equation (11-18) reduces for the case under consideration to

$$(\partial F/\partial t)_{\text{coll}} = -eE_x(\partial F/\partial p_x) \tag{11-24}$$

We shall now assume that there exists a relaxation time τ such that

$$(\partial F/\partial t)_{\text{coll}} = -(F - F_0)/\tau \tag{11-25}$$

(compare 11-20). Thus, according to the last two equations,

$$(F - F_0)/\tau = eE_x\,(\partial F/\partial p_x) \simeq eE_x\,(\partial F_0/\partial p_x) \tag{11-26}$$

where the last approximation is valid for small fields so that $(F - F_0)$ is relatively small (physically speaking this assumption is equivalent with a linear dependence of I on E). Making use of the fact that the energy of the electrons is given by $\epsilon = (p_x^2 + p_y^2 + p_z^2)/2m$, one may write instead of (11-26),

$$(F - F_0)/\tau = ev_xE_x\,(\partial F_0/\partial\epsilon) \tag{11-27}$$

Substituting (11-27) into (11-23), one obtains for the current density,

$$I_x = -(2e^2/h^3)E_x\iiint v_x^2\tau(\partial F_0/\partial\epsilon)\,dp_x\,dp_y\,dp_z \tag{11-28}$$

We shall assume that τ is a function only of the energy and not of the direction of motion (compare 11-20). Since $\partial F_0/\partial\epsilon$ is also a function of ϵ alone, one may transform (11-28) into a single integral by replacing v_x^2 by $v^2/3$ and $dp_x\,dp_y\,dp_z$ by $4\pi p^2\,dp$. Expressing the integrand in terms of ϵ, one obtains

$$I_x = -\frac{16\pi e^2(2m)^{1/2}}{3h^3}E_x\int_0^\infty \epsilon^{3/2}\tau(\epsilon)\left(\frac{\partial F_0}{\partial\epsilon}\right)d\varepsilon \tag{11-29}$$

Now we have seen in Sec. 9-3 that $\partial F_0/\partial\epsilon$ has an appreciable value only in an energy range of a few kT about the Fermi level ϵ_F. To a good

approximation $\epsilon^{3/2}\tau(\epsilon)$ under the integral sign may thus be replaced by the quantity $\epsilon_F^{3/2}\tau_F$ in front of the integral. Furthermore,

$$\int_0^\infty (\partial F/\partial\epsilon)\, d\epsilon = -1$$

and if one substitutes ϵ_F from formula (9-9), one finally obtains the simple result,

$$I_x/E_x = \sigma = ne^2\tau_F/m \tag{11-30}$$

where n is the number of electrons per unit volume. It is interesting to note that although all electrons take part in the conduction mechanism only the relaxation time of the electrons at the Fermi level occurs in the conductivity. The reason for this may be explained with reference to Fig. 11-1. The full circle represents the Fermi distribution for a two-dimensional case in the absence of an external field. In the presence of a field along the x-direction, the velocity of all electrons is shifted by an amount Δv (the average drift velocity), leading to the dashed circle. It is evident that the distribution is changed only in the vicinity of the Fermi level, so that only the relaxation time of electrons near ϵ_F is of importance.

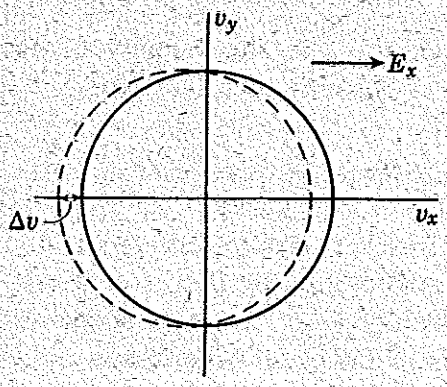


Fig. 11-1. Exaggerated representation of the influence of an electric field on the velocity distribution for a two-dimensional crystal. The fully drawn circle corresponds to the Fermi distribution in the absence of a field; the field E_x produces a shift Δv opposite to the field direction (dashed curve).

Note that (11-30) is essentially the same as (11-8), except that τ has been replaced by τ_F. Although the treatment given here was based on the free electron approximation, a similar treatment may be given for the band approximation.[6] The result of such a calculation is

$$\sigma = n_{\text{eff}}e^2\tau_F/m \tag{11-31}$$

i.e., n is replaced by the effective number of free electrons n_{eff} as defined in Sec. 10-5. It must be noted that (11-31) is based on the assumption that the energy of the electron as well as τ are functions of the absolute value of the wave vector only.

11-5. The mean free path in metals

If we confine ourselves to the conductivity of metals in the temperature region $T \gg \theta$, the existence of a relaxation time is assured according to

[6] See, for example, N. F. Mott and H. Jones, *Theory of the Properties of Metals and Alloys*, Oxford, New York, 1936, p. 258.